

OPVSim: Applying a Graph-Based Methodology for VR Training in Guided Learning of Emergency Procedures in a Ship's Engineering Room

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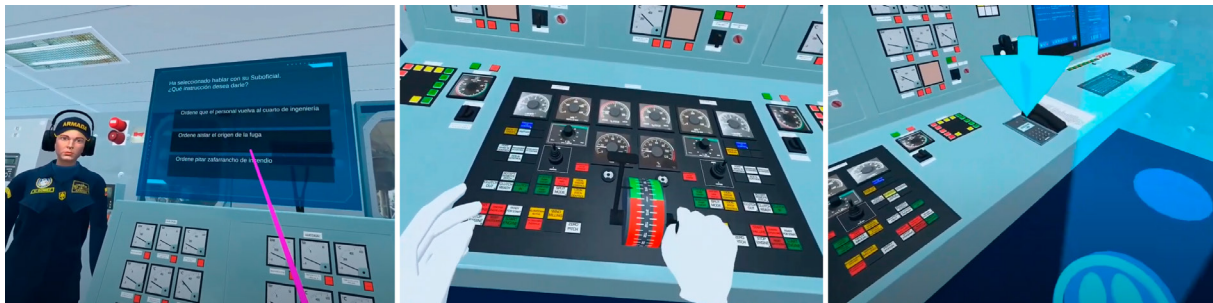


Fig. 1: VR training simulation interface showing a step-by-step emergency procedure within a ship's engineering room

Abstract—Virtual Reality (VR) can support effective and scalable training for procedures presented as step-by-step processes in machinery use, such as in an engineering room. Currently, many procedures are trained in closed experiences that do not allow for human errors or interaction with real subjects or machinery. Our goal is to address this gap by using VR training tools for guided procedural learning. To achieve this, we applied an existing methodology for designing such systems, called ProtoColVR, to develop a VR training simulation prototype for emergency procedures within an engineering room on a ship. The simulator, called OPVSim, employs end-to-end instruction, including a VR controls tutorial, step-by-step guided training based on graphs, offering different paths to the same goal, with feedback on correct and incorrect actions. We conducted a study with two groups of participants—cadets in training and experienced officers—to assess the effectiveness of the system developed using the described methodology. The study results reveal positive trends, including increased knowledge scores after using the simulator, and favorable outcomes for the prototype in terms of usability, presence, and workload. We discuss our findings, limitations, and the implications for designing VR training systems for guided procedural learning.

Index Terms—training, virtual reality, simulation, navy, militar, procedural learning, guided learning

1 INTRODUCTION

In numerous contexts, such as industrial skills training (manufacturing, maintenance, assembly) and military environments (navigation, aviation, etc.), it is crucial to train personnel in decision-making and the use of specific machinery, such as control panels. Currently, it has been observed that VR offers the opportunity to design adaptable and personalized training, which allows for efficient and effective training for a wide range of learners, thus increasing accessibility, reducing training costs, and minimizing risks for both learners and the machinery itself [11, 39]. Recent studies have demonstrated that VR is effective for training in machinery handling for manufacturing [29, 33], control panel management for aviation simulation [7], procedural training simulation [4], medical simulation [6, 18, 34] and preparing learners for real-world work environments.

In the military context, the benefit of using VR to supplement traditional hands-on training has been observed. For example, flight simulators in the U.S. Navy, such as the Aviator Training Next (ATN)

program [7], and the 2H111 simulator, the VR version of the Landing Signal Officers [15], have been implemented. These implementations were driven by the fact that traditional hands-on training approaches often present multiple limitations, such as high costs, safety risks, logistical complexity, and space-time constraints [11]. Conversely, VR provides a safe, controlled virtual environment at a relatively low cost for military personnel to practice their technical skills and improve their cognitive functions [39]. Furthermore, in the medical context, research has been conducted on the use of VR for training doctors [18]. For example, in trauma scenario simulations with military doctors, using replication and generalization for multiple cases [6], as well as simulation for training undergraduate students, both in a pre-context and during their studies [18].

While VR training systems have proven beneficial in many contexts, as previously mentioned, it is also clear that little is known about the current practices of experienced practitioners in this field, who face diverse disciplines, skills, and platforms, resulting in a fragmented environment of languages, tools, methods, and approaches [25]. Consequently, there is a clear need for methodologies or guidelines that allow creators to know where to start, design with consideration of depth, facilitate communication between multidisciplinary teams, and articulate current practices to streamline the design and implementation of VR simulators for training [14]. Although efforts have been made to define standardized methodologies, frameworks, and approaches [14, 26, 11, 28], it is necessary for the academic community to demonstrate its expertise in adapting these methods to bridge this gap and establish effective methods for simulator creation.

This article presents the development process of a VR training sim-

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ulator, using the ProtoColVR methodology [14] and a guided learning approach [26, 37] through the use of graphs. The VR simulator aims to train Engineering Watch Officers of our naval force in decision-making and control panel management within a ship's engineering room. Each procedure to be taught is designed as a simulation graph, allowing for training with immediate feedback, enabling multiple predefined paths with expected player actions (both errors and correct actions). Below, we highlight the main contributions of this article:

- Design and implementation details, explaining how the ProtoColVR methodology was applied and adapted to achieve a VR training simulator with three emergency cases for guided procedural training
- Results from the evaluation of the VR trainer for a specific case of a fuel leak, involving 16 trainee cadets and 16 experienced officers. The impact of the simulator on knowledge was assessed before and after training, demonstrating significant improvements in scores, particularly among the cadets. Additionally, the study analyzed presence and immersion in the virtual environment, revealing a high sense of presence and moderate cognitive workload, while also evaluating usability. These findings highlight the simulator's effectiveness in training and immersive learning experience.
- Lessons learned during the design and development of the VR simulator for a project with features that are generalizable to procedural training simulators.

2 RELATED WORK

2.1 Use of Virtual Reality in Military Training

Currently, VR is extensively utilized across various domains, including education and training. Reports indicate that its predominant application in organizations is for training purposes [39, 14]. This prevalence is attributable to several advantages that VR offers: reduced costs, the ability to replicate scenarios within a single virtual environment, and the flexibility to conduct training at any time and in any location. These benefits are advantageous in numerous sectors but are especially crucial in machinery and military training contexts [27]. Traditional training methods in these areas often face limitations related to high costs, logistical complexities, and spatio-temporal constraints [15, 21, 12]. Furthermore, computer simulations provide enhanced convenience, accessibility, and flexibility [12].

In this context, extensive research has focused on the use of VR for military training. Specifically, studies [15] have identified two main approaches: (1) solutions that offer more effective training by enabling faster and more durable skill acquisition while saving human and material resources compared to traditional methods, and (2) VR solutions that are the only viable means to acquire certain skills (e.g., pilots practicing emergency procedures). Accordingly, the literature includes VR simulators for military training in procedures [15, 8], machine and equipment operation [2, 3, 7], and decision-making [12, 19, 22, 7].

These studies have provided various insights, approaches, and contributions regarding the use of VR for military training [39]. For instance, the calibration system developed by Taupiac et al. [36] for the FELIN equipment has significantly improved learning efficiency and soldier motivation, providing a more immersive and realistic VR training environment. Similarly, Doneda and de Oliveira [8] developed the Helicopter Visual Signal Simulator (HVSS), offering a cost-effective and efficient solution for helicopter visual signaling training compared to traditional methods. Bailey, showed a U.S. Navy VR simulator for training maintenance procedures [2]. Greunke and Sadagic [15] designed a VR training system that uses commercial devices to complement the traditional Landing Signal Officer Trainer, Device 2H111, enhancing its functionality and reception. Furthermore, the Aviator Training Next (ATN) program has demonstrated that VR can train pilots with competencies comparable to those trained on actual aircraft [7]. Hussain et al. [22] also contributed to this field with the development of the Flooding Control Trainer (FTC), used to train new recruits in essential skills for the U.S. Navy. These VR innovations are providing more accessible and effective solutions, addressing many of the limitations of traditional training equipment and methods [39].

2.2 Standards and methodologies for the creation of VR simulators for training

While it is crucial to understand the development and creation of applications and simulators for training purposes, it is equally important to establish effective methods, guidelines, and frameworks for building such systems in a way that yields positive results. In particular, considering the multitude of factors involved, from the design phase (capturing requirements, understanding the problem context with multidisciplinary teams), to development (technical aspects, incremental implementation with prototypes of varying fidelity), and finally to testing these prototypes (validation with experts at each stage of creation). However, to the best of our knowledge, recent research has shown that little is known about the current practices of experienced practitioners in this field. These practitioners face challenges due to the diversity of disciplines, skills, and platforms, leading to a fragmented landscape of languages, tools, methods, and approaches [25, 14]. Consequently, recent efforts have been made to bridge this gap. Specifically, in the training context, notable research includes the IME^{VR} framework [11] and the ProtoColVR methodology [14].

On one hand, IME^{VR} is a framework designed to facilitate the development of military training simulators in virtual reality. It provides an MVC structure aimed at enabling developers to focus on specific elements, such as user interface and 3D interaction, rather than the fundamental functionalities of a VR application. During the design phase, key considerations included performance evaluation by instructors, integration of VR devices (HMDs), network communication between terminals, and performance measurement. IME^{VR} thus offers a flexible structure that reduces development effort and optimizes the creation time for military simulators by providing predefined functions for real-time interaction and monitoring [11]. Additionally, an accompanying paper details a case study of a simulator that inspired the development of this framework [12].

On the other hand, ProtoColVR is a methodology that outlines a series of stages and steps for working with multidisciplinary teams in the creation of VR training simulators, particularly focused on procedural training. It describes how to create prototypes of different fidelities and test them with experts using incremental virtual environments and simulation graphs. Moreover, it includes a Unity plugin called Tweenity [13], which facilitates the integration of artifacts such as virtual environments and the graphs that describe each training procedure. This methodology was tested during a hackathon-style event and applied to two real simulators [14].

2.3 Procedural Guided Learning Approach

It has been demonstrated that VR environments facilitate learning and skill development in both cognitive and psychomotor domains. In particular, due to their sense of presence [9] in a computer-generated virtual world, as if it were a real-life setting, these unique immersive characteristics can increase learners' affective arousal and enhance their cognitive processing [17, 32]. Additionally, it has been found that HMDs are useful for skill acquisition. These include cognitive skills related to remembering and understanding spatial and visual information and knowledge; psychomotor skills related to head movement, such as visual scanning or observational skills; and affective skills related to managing emotional responses to stressful or challenging situations [24].

In the context of learning, research has identified various types of learning, particularly within educational, military, and sports settings [1, 10], including observational, social, and procedural learning [31, 4, 14]. The latter involves learning sequences of steps or instructional activities that require students to perform specific tasks. This type of learning has been shown to be particularly effective, as it allows students to engage in activities related to high-risk topics without exposing themselves or others to real-world dangers. Additionally, it facilitates repeated practice sessions [5].

Some studies have found that procedural learning can be enhanced by using guided learning environments [37], which provide step-by-step guidance, including object highlights, scoring, and visual and auditory messages. Additionally, other research [26, 35] suggests of-

fering greater freedom in the learning process, allowing students to achieve the same goal through different paths, again with visual aids and immediate feedback. This approach has been found to support the constant practice learning strategy, which, according to research [23], may be more beneficial for novices compared to experienced learners.

Based on the above, we decided to implement a guided procedural learning system to teach the three procedures described in Section 3.2. This system was specifically designed to ensure that participants could learn the necessary steps and interactions required with the control panel in the ship's engineering room, as well as the instructions that need to be communicated to the crew. By leveraging the ProtoCoVR methodology, we demonstrated practical applications of current approaches, using graphs to structure both the step-by-step process and the feedback necessary for real-time guidance during training. Furthermore, in line with the findings of [26, 35], we used graphs to define multiple learning paths, allowing participants to reach the same goal through different routes, thus promoting flexibility and adaptability in the learning process. This approach aims to showcase the creation of the simulator, applying current techniques and offering a use case that can serve the community in developing effective simulators aligned with characteristics such as the procedural learning.

3 SYSTEM DESIGN

To design the VR simulator, we decided to adopt the ProtocolVR methodology [14], considering that the characteristics of our project aligned with the features of generic training projects described in it:

1. **Specific Audience:** The training was aimed at the Engineering Watch Officers (EWOs) of our naval force, who are responsible for attending to and making decisions in the engineering room during emergencies.
2. **Remote Team:** The institutions involved in the development of the simulator, our educational institution and the naval academy, were located in different cities. Therefore, requirements had to be gathered remotely, both for defining the simulation steps and for specifying the virtual environment.
3. **Multidisciplinary Team:** The project involved a multidisciplinary team consisting of 2 developers, 6 naval officers, and 3 designers.
4. **Real Environment to Virtualize:** The simulator environment would result from "virtualizing" the engineering room of a naval vessel in our fleet.
5. **Procedure or Series of Steps to Train:** The VR simulator aimed to provide guided training for 3 cases of Emergency Reaction Movements (known as MOB exercises), which involve steps for managing a control panel and decision-making with the crew.

Additionally, given these characteristics, the proposed methodology was well-suited to our project, as the prototypes it recommends creating are based on the environment to be virtualized and the definition of the steps of the procedure to be trained (in the form of graphs used both in design and development). In this section, we will explain how this methodology was adapted to the specific case of our project, the decisions made, and the results obtained in the design and creation of the simulator.

3.1 Virtual Environment

To ensure the effectiveness of the prototype, one of the primary considerations was to recreate a training environment as closely resembling the real one as possible. Traditional hands-on training involved simulating emergencies within the engineering room of a real vessel, with no interaction allowed with the machinery or control panel due to potential risks. In this context, the use of VR was proposed to design a virtual environment that accurately replicated the appearance of the engineering room of an Offshore Patrol Vessel (OPV). This approach aimed to maintain familiarity with the real environment while allowing full interaction with its elements.

To achieve this goal, the "Stimulators" and "Demonstrators" prototyping approach [14] was adopted, using 3D online conferencing tools

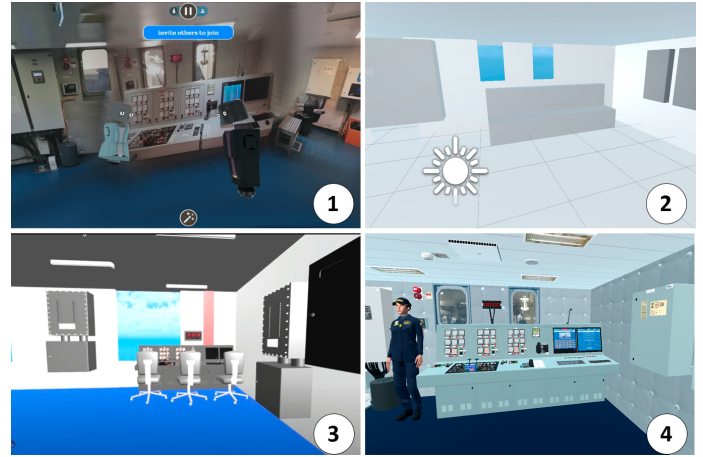


Fig. 2: Scene evolution

to facilitate discussions with experts and quickly define training objectives, understand the real environment, and determine interactions with objects for effective training. Additionally, traditional meetings were conducted, and asynchronous materials, such as videos of real procedures, were shared to gather information and observe these procedures in order to carry out the Task Analysis.

Consequently, the project began with the capture of "spatial requirements," which involved a detailed identification of visual elements in the VR simulator environment, such as the dimensions of the real space, photographs of the location, elements to be modeled in 3D, and those requiring interactivity. To this end, three early low-fidelity prototypes were developed (see items 1-3 in Figure 2), focusing solely on the discussion and validation of these elements.

3.1.1 Skybox with Photographs of the Real Vessel

To start, given the need to understand the real space and how interactions were carried out within it, a skybox was created using photographs of the control room of a vessel (see item 1 in Figure 2). This skybox was integrated into a 3D meeting creation tool, which facilitated discussions with experts about the layout of the real space, the relevant elements to model, and the definition of their locations and interactivity.

In addition to this activity, a designer created blueprints of the real environment to assist in the development of the 3D model of the engineering console and the basic structure of the room (walls, stairs, tables, and panels).

3.1.2 Grayboxing Prototype of the Engineering Room

Based on the results obtained from the virtual session, the development of the first 3D prototype (see item 2 in Figure 2) proceeded using the grayboxing technique, which involved creating a 3D "mockup" [14] of the space. Subsequently, another meeting was organized within this environment, again in a virtual space, to discuss with domain experts the layout of the 3D environment, the available elements, their potential interactivity, and their scale and location.

3.1.3 3D Prototype of the Engineering Room

Finally, improvements derived from the virtual sessions within the Grayboxing prototype were implemented to create a new environment featuring textures, 3D elements, and additional details (see item 3 in Figure 2). Six iterations of this scene were conducted, again through meetings with experts, until a version was achieved that accurately reflected the engineering room to be virtualized. In the third iteration of the 3D environment, a computer prototype was developed to evaluate the sequence of steps with a domain expert. After refining this prototype, the final version intended for the VR environment was generated (see item 4 in Figure 2).

3.2 Trainig Procedures

3.2.1 Training Cases

Training in MOB exercises was crucial due to the primarily theoretical nature of the doctrine. Cadets had a written manual for study, and practice involved "acting out" the situations aboard the ship, without the opportunity to interact with real machinery. Therefore, implementing a VR solution offered the potential to significantly enhance training by allowing cadets to learn and practice in a virtual environment that faithfully replicates the engineering room, where interaction with the machinery can occur without risking damage to actual equipment, as it is a virtual version. In particular, since there are many MOB emergency exercises, after consulting with expert officers, three cases were selected for training through the simulator. These cases required interaction with machinery and represented the primary types of emergencies to address:

1. Fuel leak in a propulsion engine
2. Unusual noises or vibrations in the reduction gear
3. Overheating in a bearing of the main shafts

3.2.2 Simulation graphs

For each case, it was necessary to conduct a thorough requirements gathering process, as well as to define the specific tasks and steps to be trained. Each step needed to be clearly defined in terms of: the conditions under which it would occur, how it would be visualized in the engineering room, the task or action to be performed, and the consequence or system response.

To achieve this definition, we perform a Task Analysis (as proposed in ProtoColVR) and describe the results in the Twine graph system with the Tweenity format employed [14, 13]. This approach allowed each step of the procedure to be represented as a node, capturing the aforementioned characteristics.

The primary goal of using simulation graphs was to clearly establish the steps of the MOB exercises to be simulated and to define the specific tasks that the player, in their role as EWO, needed to perform. While other developments have reported the use of flow diagrams or task lists [37, 26, 8, 26], we considered that the graphs from the ProtoColVR methodology were more suitable for our project's needs, due to the following benefits:

- Having a simple format to define the simulation steps in collaboration with the domain expert officer.
- Ability to define how to provide feedback on errors at each step and use elements that enhance guided learning (such as highlighting objects, playing audio cues, etc.).
- Ability to define different paths to model more possibilities in the procedure to be trained, so that during guided training, feedback can be provided for the most common error cases.
- Ability to reuse the same artifact created during the step definition phase to control the execution of the simulator, using the Tweenity plugin [13].

Initially, expert officers were asked to create a preliminary version of the graph in natural language. This preliminary version was then subjected to collaborative sessions with the entire multidisciplinary team, where the graph was reviewed and refined. During these sessions, potential errors that could occur at each step of the procedure were identified, which allowed for the creation of intermediate steps designed to provide feedback and guide the learner towards the correct option. In particular, it was especially interesting because the expert could "simulate" the situation using a special "play mode" of the graph, which allowed us to analyze the tasks and steps, in order to iterate and improve each simulation graph.

This process was crucial for gaining an in-depth understanding of the role to be trained and for defining the types of actions and decisions that the EWO performs and, therefore, needed to be integrated into the training:

- Determine the interactions to be performed with the machinery (e.g., buttons to press, handling of levers, operating screens showing engine status).

- Identify the commands that can be given to the watch personnel (given that the EWO oversees three petty officers: the Watch Petty Officer, the Propulsion Petty Officer, and the Electrical and Refrigeration Petty Officer).
- Learn the order of the procedure according to emergency conditions, starting with primary tasks (for the safety of the ship and crew) and then addressing secondary tasks (preventive actions).

Thus, for each step of the procedure, the expected action of the EWO was explicitly specified, as well as how the system would correctly respond to that action or provide feedback if it was not performed properly. For instance, in the case of a fuel leak in a propulsion engine, one of the steps involves assessing the amount of spilled fuel. If the amount exceeds 10 gallons, a fire drill must be activated; if it is less, a preventive procedure must be followed. Figure 3 illustrates this specific part of the graph.

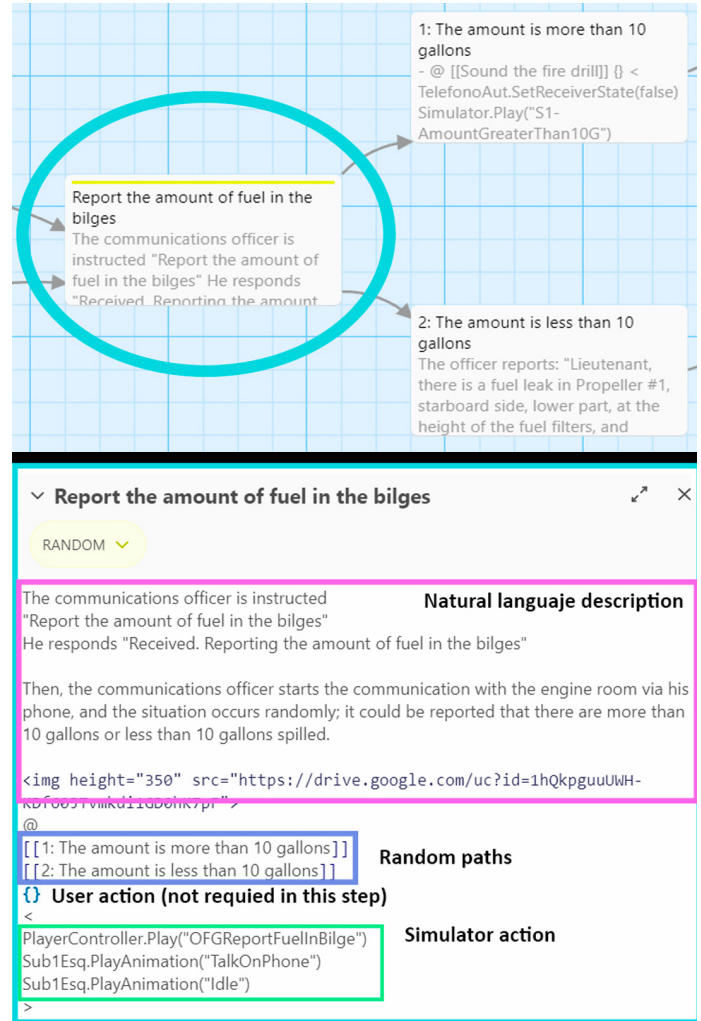


Fig. 3: Step Definition: Description, Expected User Action/Interaction, Simulator Response

The design of each graph was carried out through a combination of asynchronous editing by experts and a series of synchronous collaborative sessions, until the final version was achieved. Once completed, each graph contained between 50 and 60 nodes, allowing for the representation of nonlinear procedures, where paths varied according to case conditions or feedback generated when modeling error scenarios. Figure 4 shows part of the graph corresponding to the fuel leak case in the propulsion engine.

To design the graphs, the node types defined in ProtoColVR [14] and Tweenity [13] were utilized. These allowed for the modeling of each step's behavior to establish whether one of several action options

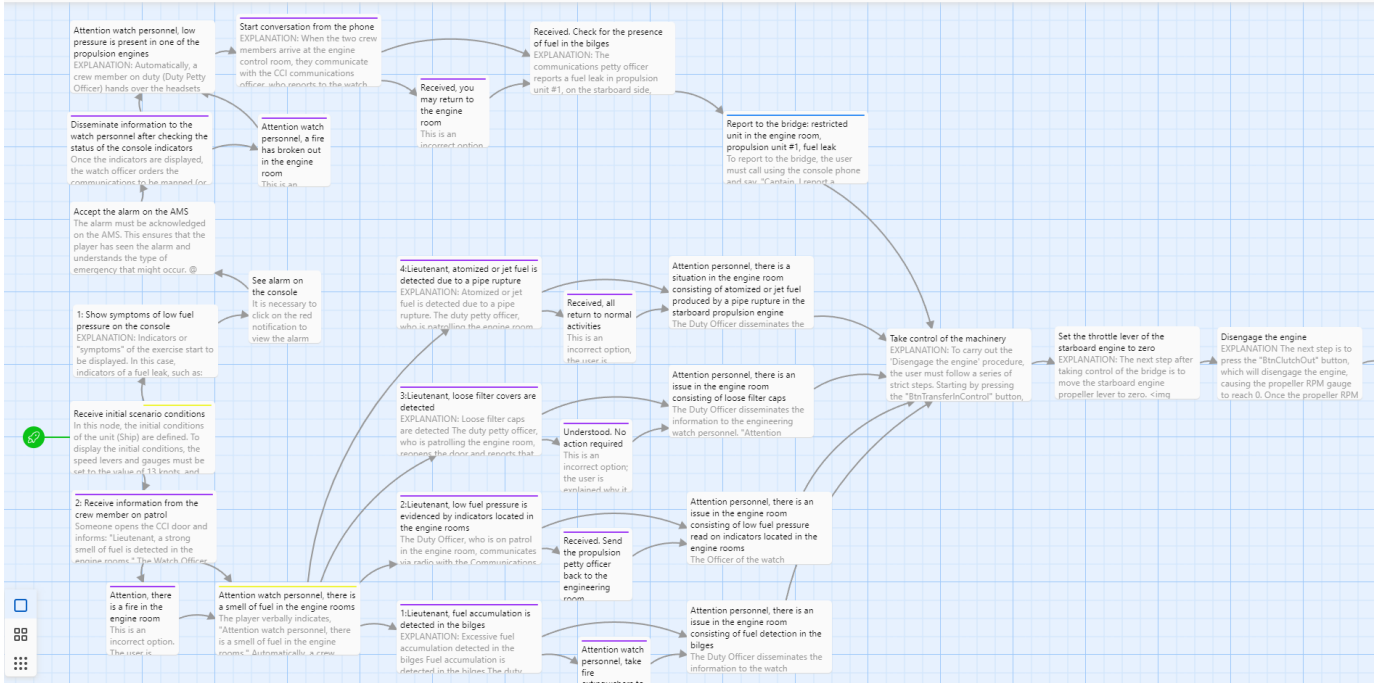


Fig. 4: Graph: steps (part) of the complete training procedure of Fuel Leak available at [Supplemental Materials](#)

was expected (multiple choice node), if there was a wait time at the step (timeout node), or if paths would be defined randomly (random node). For example, in a step, alarm conditions could be presented in two ways: one is a petty officer entering to report a fuel smell in the engine room, and the other is a low-pressure alarm on a propulsion engine from a control panel monitor. To simulate this behavior, a "random node" was used to randomly trigger one of the two situations.

4 SYSTEM ARCHITECTURE, DEVELOPMENT AND PROGRAMMING ENVIRONMENT

The system was built using an HP 16.1" gaming laptop, equipped with an Intel Core i7-10750H CPU @ 2.80 GHz, 12GB of RAM, and a NVIDIA® GeForce RTX™ 2060 GPU. The system was specifically designed for use with PC-VR and was tested on both the Rift S and Meta Quest 2 using Oculus Link. This choice was influenced by the availability of 10 Oculus Rift S headsets at the Naval School, which were intended to be used for training.

The development and programming environment consisted of the Unity game engine (version 2021.1.13f1), Twine for creating graphs, Blender for model editing, Photoshop and Substance for texture manipulation, and Audacity as the audio editor. The deployment of the final executable was facilitated by the Oculus XR Plugin. Input integration and object interactions were implemented using the Oculus Integration Plugin. The Tweenity plugin was used to integrate and execute simulation graphs and provide immediate feedback.

The sounds for the engineering room were created using recordings from the real space, including alarms, button sounds, ambient noise, and more. This was done with the intention of enhancing the sense of presence and, consequently, improving learning outcomes.

4.0.1 Tweenity integration

Upon importing the Tweenity plugin [13], a series of game objects were obtained that allowed the graph to be read within Unity for simulation control. To achieve this, the graphs written in natural language, developed in collaboration with experts, were incorporated along with a scripting section to establish the expected interactions, the game objects to be interacted with, and the simulator's responses, all in terms of invoking the functions implemented by our development team. For example, as shown in Figure 5, in one step, the Engineering Watch Officer (EWO) is required to take control of the machinery. The ex-

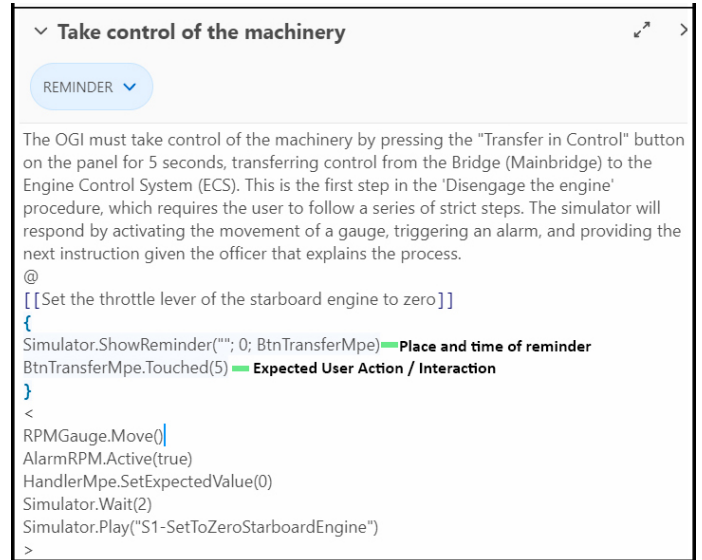


Fig. 5: Sample step with Tweenity: shows the node in natural language and then in scripting

pected action is to press the "Transfer in Control" button on the control panel for 5 seconds, after which the simulator responds by activating a pressure gauge, an alarm, and the next instruction given by the human guide that instructs the process.

To use the graphs created in Twine during the simulation, the instructions from the plugin repository were followed for the correct export of the graph in text format. This format needed to be interpretable by Tweenity plugin inside Unity. This process presented some challenges, which will be described in Section 7.

4.0.2 Features

The VR system consists of three components: a VR tutorial to familiarize users with VR interactions, a procedure selection module for choosing the specific training scenario, and a guided training session, where the learner initiates the simulation of the procedure to be trained.



Fig. 6: Step-by-step guidance elements

Training Procedure Selection When starting the simulator, the user is first given the option to either complete a VR tutorial or proceed directly to the simulation. If the user chooses to enter the simulation, three training scenarios, described in Section 3.2, are presented. After selecting one of these scenarios, the user can engage in training through a linear procedure simulation or explore the training of a procedure with multiple pathways, which are designed based on different conditions and steps that lead to the completion of the same task.

VR tutorial This stage enables users to practice and familiarize themselves with the controller operations within the simulation. Users learn fundamental actions such as hand gestures, object manipulation, including grabbing and releasing items, interacting with specific control panel objects like buttons and levers, learning how to teleport, and interacting with the user interface. This brief tutorial, spanning a few minutes, equips users with the essential VR proficiency necessary for the training sessions.

Guided Training session During the training session, one of the procedures is simulated. To achieve this, the graph previously designed with experts is executed, activating alarms or audio cues from the engineering staff to simulate the situation and allowing the trainee to perform interactions as required by the step (e.g., pressing buttons, operating levers, making calls, speaking with personnel in the control room, interacting with the control panel system). Throughout the process, a step-by-step guidance is provided, offering a structured approach to learning. As shown in Figure 6, this includes object highlights using arrows, marks and outlines, timeouts to show the highlights, text feedback, and audio messages through a human guide who explains the procedure.

5 STUDY DESIGN

This section describes the study participants, study procedure and measurements.

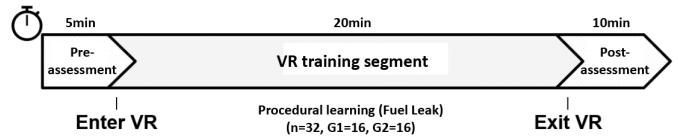


Fig. 7: Flowchart of the user test

5.0.1 Participants

The test was conducted with two groups of participants: The first group (G1) consisted of experienced naval officers (Lieutenants), and the second group (G2) comprised cadets in training. Participation was voluntary, motivated by the participants' interest in exploring the VR simulator.

All participants were over the age of 18, in accordance with the legal definition of adulthood in our country. No special requirements were imposed, other than belonging to one of the two designated groups.

5.1 About Institutional Review Board Approval

In the context of testing the VR system designed for Navy officers' training, the current protocols and regulations of the military institution require no additional ethical approval from an ethics committee, provided that the VR system poses no significant physical or psychological risks to participants, as the testing was conducted in a controlled and safe simulated environment. This study adhered to the current guidelines of both the military institution and our university. Therefore, all participants voluntarily provided informed consent, fully understanding the purpose and procedures of the testing. The data collected were anonymous and used solely for system evaluation and training improvement, with no identifiable information involved.

5.2 Study Set Up and Measurements

The goal of this study was to evaluate the effectiveness and impact of the guided procedural VR training system developed using the Pro-toColVR methodology. This evaluation was conducted in terms of knowledge acquisition, perceived workload, and system usability. To achieve this, training performance was assessed using six assessments:

1. **Participant demographics (PD):** This assessment collected information on age, gender, current experience, and group (cadet or officer).
2. **Pre-training knowledge assessment (PreTK):** This assessment measured participants' understanding of the key steps involved in handling a fuel leak in a propulsion engine before the training.
3. **The Presence Questionnaire (PQ):** This questionnaire, developed by Witmer and Singer [38], in its reduced version of 20 questions, assesses the degree to which direct interaction with a VR environment makes participants feel that the simulation sensations are real [9].
4. **The NASA Task Load Index (NASA-TLX):** This index [20] was used to measure participants' perceived workload during the training.
5. **Post-training knowledge assessment (PostTK):** The same PreTK, but applied after using the VR system.
6. **Usability questionnaire using the Single Ease Question (SEQ):** This type of questionnaire, using the SEQ method [16], gathered feedback on the usability of the core functionalities of the VR system.

All our assessments with detailed questions are available in the [Supplemental Materials](#).

5.3 Procedure

The study was conducted in person at a naval academy of our national navy over a two-day period, with one day allocated for each group of participants, in sessions averaging 35 minutes. As illustrated in Figure 7, the study comprised a pre-assessment, a VR training segment with guided procedural learning on handling a fuel leak in a propulsion



Fig. 8: Main interactions in OPVSim

engine (with 24 steps), and a post-assessment. . All training segments were recorded via screen capture.

At the start of the session, participants were introduced to the study and provided with informed consent to confirm that they had been properly briefed and had agreed to participate. This consent form included each participant's information and assigned a user code to anonymize the assessment responses for later analysis, using a format of 'G[groupNumber] - [userNumber]' (e.g., the first user of group 1 is G1-01).

In the pre-assessment activity, participants completed a demographic survey and a pre-training knowledge assessment (PD and PreTK). The knowledge assessment consisted of five questions concerning key actions required in a case of fuel leakage in a propulsion engine. It included one single-choice question, two multiple-choice questions (where a score was assigned to each selected option, dividing the total points by the number of correct options), and two "yes" or "no" questions. All questions also included an additional "I don't know" choice to allow participants to indicate if they had no prior knowledge. This assessment resulted in a score ranging from 0 to 5, where 0 is the lowest score and 5 is the highest.

Subsequently, guided training was conducted using the VR simulator, focusing solely on one of the exercises: handling a fuel leak in a propulsion engine, to ensure greater uniformity in the test. Additionally, variability was reduced to a single possible path, with minor variations in the steps for issuing commands to the personnel, totaling 24 steps to be trained.

Following the training segment, participants completed a post-assessment to evaluate their sense of presence, task load, post-training knowledge, usability, and general feedback using the PQ, NASA-TLX, PostTK, and SEQ questionnaires. The 19-item Presence Questionnaire (PQ) [38] was used to gauge participants' feelings of presence, or "sense of being there," within the training environment. Responses were rated on a scale from 1 ("Not at All") to 7 ("Completely"), reflecting the degree of agreement with each item. The NASA-TLX [20] employed 7-point Likert scales to assess mental demand, physical demand, temporal demand, performance demand, effort, and frustration. The PostTK questionnaire included the same questions as the PreTK to compare knowledge acquired after using the VR simulator. Finally, the usability questionnaire evaluated the ease of use of eight main interactions (see Figure 8) using the Single Ease Question (SEQ) [16] method, where 1 indicated "very difficult" and 7 indicated "very easy."

6 RESULTS AND ANALYSIS

We examined demographic, knowledge, presence, and usability outcomes, in addition to perceived workload. Each analysis focused on comparisons between pre- and post-VR-training measurements. All analyses were conducted using IBM SPSS.

6.1 Demographics

The study involved 32 participants ($n=32$) divided into two groups. G1 consisted of 16 naval officers (Lieutenants), all of whom had prior experience in navigation and staying aboard a ship. G2 comprised 16 cadets in training, the majority of whom had previously visited ships (10 out of 12). In terms of age distribution, 50% of the participants

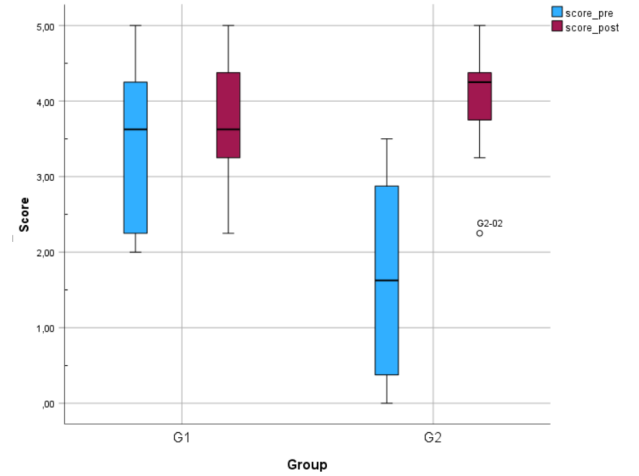


Fig. 9: Pre and Post Scores in G1 and G2

were aged 18 to 24, all from G1; 37.5% were aged 25 to 34; and 12.5% were aged 35 to 44. The majority of participants, 93.8% ($n=30$, $G1=14$, $G2=16$), were male, while 6.3% ($n=2$, $G1=2$, $G2=0$) were female, due to the limited availability of females at the time of the study, who met the characteristics of the described groups, which may be related to the lower enrollment of women in the Naval Academy, where only 20% of admitted students in 2024 were female [30]. Regarding prior knowledge, 68.8% of participants indicated that they were not familiar with the MOB doctrine for handling a fuel leak in a propulsion engine. This included 100% (16) of G1 and 37.5% (6) of G2. Additionally, 18.8% (6) of participants (12% of G1 and 25% of G2) had previous experience using VR, while 81.3% did not.

6.2 Knowledge

We constructed Shapiro-Wilk tests and Q-Q plots to examine the normality of the data ($p>0.05$). The results indicated that the knowledge score metrics were normally distributed both PreTK and PostTK, in the whole group and in G1 and G2 separately. Descriptive statistics and p-values are shown in Table 1.

Table 1: Normality and Descriptive Statistics of Pre and Post Scores for the Complete Group, G1, and G2

	Complete Group		G1		G2	
	Pre	Post	Pre	Post	Pre	Post
p-value	0.22	0.15	0.06	0.60	0.10	0.84
Mean	2.48	3.91	3.41	3.78	1.56	4.03
Median	2.38	4.00	3.62	3.63	1.63	4.25
σ	1.45	0.73	1.01	0.79	1.23	0.65

For these results, an average improvement of 1.43 points in the test scores can be observed ($Mean^{pre} = 2.48$, $Mean^{post} = 3.91$). In total, 75% of the participants (24/32) increased their scores. Additionally, although an increase in scores was observed in both groups (see Figure 10), this increase was much more evident for G2 than for G1, which was expected, as G2 was composed of cadets in training with no prior experience in these types of exercises. This led to a larger difference initially, but it showed that the simulator helped them learn the main steps for handling the emergency. In particular, in G1, 100% of the participants improved their scores, with an average score increase of 2.62 points and a higher average post-score ($Mean^{post-G2}=4.25$) compared to the experienced officers ($Mean^{post-G1}=4.03$).

Since the scores corresponded to independent subjects, with two measurements taken for each at different points in time (pre- and post-VR training), and these followed a normal distribution, we decided to perform a paired t-test to determine whether the difference in knowledge scores before and after using the VR simulator was significant.

Table 2: p-values from the Paired t-test for the Complete Group, G1, and G2

	Complete	G1	G2
p-value	<.001	0.154	<.001

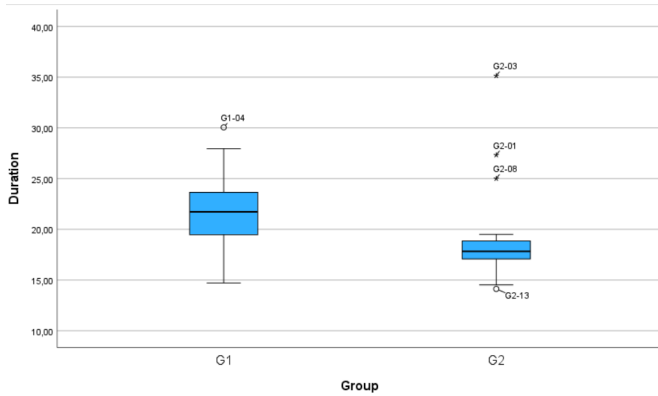


Fig. 10: Duration of the training segment

As can be seen in Table 2, the difference is statistically significant ($p < 0.05$) for the results of the Complete Group and specifically for G2, where a significant increase in the knowledge score was observed. This demonstrates the effectiveness of the simulator for guided procedural learning in cadets in training.

6.3 Training time

In terms of time, as observed in Table 3, the average duration of the training segment was 20.62 minutes for the entire group. When comparing the groups separately, no significant differences in average duration were noted. However, five participants were identified as outliers, with durations equal to or exceeding 25 minutes. When relating these durations to their knowledge scores, three of these participants achieved acceptable results (post-score ≥ 3.25), while two achieved very good results (4.25 and 5.0). None of these participants had prior VR experience, which may be related to their extended duration.

Table 3: Descriptive Training Time Statistics

	Complete Group	G1	G2
Mean	20.62	21.80	19.45
Median	19.34	21.72	17.83
σ	4.75	3.88	5.35

6.4 Perceived workload, presence and usability

Considering the results from Table 4, the NASA-TLX analysis suggests that participants experienced a low-moderate workload, with an average total score of 17.38 (CG=17.38, G1=17.00, G2=17.75) out of a maximum of 42. The reliability of the scale, measured by Cronbach's alpha, was found to be good for both G1 ($\alpha=0.81$) and G2 ($\alpha=0.78$) when analyzed separately. In particular, the dimension with the highest perceived workload was mental demand, both in the complete group and in the separate analyses of G1 and G2, which aligns with the cognitive demands of the learning task. In contrast, frustration was the dimension with the lowest perceived workload, suggesting that participants did not experience significant levels of frustration. Other dimensions, such as physical demand, temporal demand, performance, and effort, varied in their scores; however, the overall results indicate that while the task was cognitively challenging, it was neither particularly stressful nor physically demanding.

In terms of presence, as observed in Table, the average score of 102.22 (out of 133) indicates a high sense of presence among participants. Indeed, 81% ($n=26$) of participants reported a high sense

of presence, with acceptable reliability according to Cronbach's alpha ($\alpha=0.68$).

Table 4: Mean Statistics for NASATLX Dimensions and Total Score

	Mental	Physical	Temporal	Performance	Effort	Frustration	Total score
Complete	4.41	2.47	3.72	2.13	3.16	1.50	17.38
G1	4.31	2.81	3.38	2.00	3.06	1.44	17.00
G2	4.50	2.13	4.06	2.25	3.25	1.56	17.75

Table 5: Presence Score Distribution by Range

Range	Complete (n=32)	G1 (n=16)	G2 (n=16)
Low (19-56)	0% (0)	0% (0)	0% (0)
Medium (57-94)	38% (6)	13% (2)	25% (4)
High (95-133)	81% (26)	88% (14)	75% (12)

Table 6: Mean Usability Statistics for Interaction Tasks

	Hand Menu	Buttons	Levers	Teleport	Phone	Grab Tags	Dialogues	System Screen
Mean	6.09	6.09	4.09	6.56	6.06	5.44	6.47	6.06

Finally, as shown in Table 6, in terms of usability, the SEQ questionnaire achieved an average score of 46.9 (out of 56), with high reliability according to Cronbach's Alpha ($\alpha=0.86$). This indicates that the usability results were favorable, suggesting that the system was perceived as easy to use. Interactions with buttons, menus, teleportation, phone, and system screens received scores above 6. In contrast, the interaction with the lowest score was the use of levers (mean = 4.09); open comments revealed that participants did not perceive the movement as realistic, particularly in terms of speed. Conversely, the handling of dialogues and instructions for the staff received the highest score (mean = 6.56). Comments indicated that providing feedback to the staff was crucial, and the use of dialogues was executed effectively.

In summary, the analysis of the VR training simulator results shows a significant improvement in participants' knowledge after training, with an average increase of 1.43 (in scale 1 to 5) points in test scores. Cadets (G2) demonstrated a greater improvement compared to experienced officers (G1), with an average increase of 2.62, which aligns with the cadets' lack of prior experience with such simulations. The average training time was 20.62 minutes, with some participants achieving excellent results after extended periods of training. Perceived workload was low-moderate, with a high mental demand reported, while the sense of presence was high, indicating effective immersion in the simulator. Usability ratings were favorable, although interaction with the levers received the lowest score due to perceived unrealistic movement. Overall, the simulator proved effective for guided procedural training, particularly for the cadets in training.

7 LESSONS LEARNED USING PROTOCOLVR

The use of the ProtoColVR methodology enabled us to develop a VR simulator for procedural training based on guided learning. According to our study, the results were positive in terms of acquired knowl-

edge, usability, immersion, and low workload, with an intermediate-high level of mental workload. Throughout the VR simulator creation process, we gathered a series of valuable lessons that may be useful to the academic community in the development of simulators with similar characteristics.

Given that the use of VR for training has been evidenced in industry [39], in academia, and in general in the fields of VR development and creation, it is crucial to have guidelines aligned with educational needs and the creation process in multidisciplinary teams. When using ProtoColVR as an alternative to these guidelines, it is essential to consider the technical learning curve that may arise from employing tools such as Metaverses for requirements gathering and problem understanding. This approach allows for effective information collection (evidenced by high immersion scores), early results from the technology, and facilitates understanding both from the expert to the technician (through natural language graphs and Skybox Metaverse meetings) and from the technician to the expert (by presenting progress through graphs loaded into the simulator and enhanced 3D scenarios).

However, not all experts are familiar with the technology; therefore, prior training is necessary to facilitate access to the selected Metaverses for capturing "spatial requirements" and using Twine, especially during asynchronous edits. In our case, we developed written manuals in addition to conducting an initial joint editing session. Ideally, an artificial intelligence tool for generating transcriptions of meetings in Metaverses would be beneficial; however, we recorded the sessions and took notes during and after each meeting. Although this is not outlined in ProtoColVR, it can support project development and understanding of requirements.

To the best of our knowledge, there is no detailed information on capturing such scenario or spatial requirements in related work searches. However, we consider it to be of great importance and it was vital for this project, given the virtualization of the real environment. As observed in the results, this allowed for greater immersion, which has been associated with improved learning outcomes [37, 5, 17, 32].

During sessions in the virtualized environment (inside Metaverses), we observed that capturing detailed information is crucial. The preparation for the session can focus on:

- Defining which objects are important for the simulation and their behavior.
- Discussing the dimensions and appearance of the scene objects.
- Analyzing the spatial distribution of the environment and its elements.
- Identifying irrelevant or missing objects.
- Reviewing the behavior and significance of each object in the simulation.

In terms of graph creation, since our approach was guided learning, detailing error cases and feedback nodes was crucial for enhancing learning. For projects adopting this approach, we suggest conducting several collaborative sessions to define the graphs, as asynchronous edits may lead experts to omit what they consider "obvious" cases. When using ProtocolVR, the graph-building process should be oriented as a discussion with the expert, where a non-expert tries to understand how each operation is performed and the technical language involved. For example, in one case, one of the steps described by experts was "the engine must be disengaged," but this action of "disengaging" involved at least six additional steps. We consider that defining these steps through a graph is an effective method, although it is an iterative process requiring in-person or virtual interaction with the subject matter expert.

Regarding the use of tools, we encountered difficulties when adding images to each node, as they were only visible in the local version of Twine. Integration with Tweenity presented technical details that need improvement when loading the graph. It is advisable to read the plugin documentation, though ideally, a graph editor within Unity would be beneficial. If changes are required, they must be made in Twine and then exported, which involves multiple transitions between tools.

However, focusing development on interactions rather than narrative was facilitated by the plugin, as the entire procedure was outlined

in the graph, and it was sufficient to load it into Unity to execute the simulation.

We consider that improving such tools is possible by integrating an event tracking system for better monitoring of learners, the ability to combine nodes such as multiple-choice (allowing multiple user interactions) with timeout nodes (maximum waiting time for an interaction before the simulator responds) [14], and enabling graph editing directly within Unity.

8 LIMITATIONS AND FUTURE RESEARCH

This study faced several limitations. Firstly, the small sample size ($n=32$), constrained by the challenges associated with the availability of experienced navy officers, may affect the robustness of the statistical results. Although the findings appeared statistically significant, they should be interpreted with caution. Secondly, the varied levels of VR experience among participants influenced their interaction with the system. Some participants, who were new to VR, required additional time and effort to familiarize themselves with the basic operations and controls, which may have impacted their learning experience. A third limitation was the absence of comparisons with other learning methods, such as a control group or groups utilizing the current hands-on method. This limitation arose from the logistical constraints related to the need for vessels to conduct a comparison with a controlled sample, which proved difficult to manage within our Navy.

Future work should address these limitations by evaluating a larger sample and incorporating additional cases without aids to assess the training through automated practical assessments utilizing graphs. Specifically, we plan to develop graphs for varying levels of difficulty. This would involve removing the aids from the graph (e.g., reminder nodes) and modifying simulator actions so that feedback reflects a penalty or the actual consequence of incorrect actions. Such an approach would facilitate the practice of learned skills through the use of graphs. Furthermore, we aim to implement a learning tracking system that monitors the different paths taken by learners and an automated evaluation system that assigns weights or scores to nodes and paths taken to achieve the final goal.

9 CONCLUSIONS

This paper explores the design and development of a VR simulator for guided learning in emergency procedures within a Ship's engineering room. The system was developed by a multidisciplinary team that included developers, designers, and domain experts. We employed the ProtocolVR methodology for its implementation, aiming to serve as a case study on applying existing methodologies to address the lack of guidelines and standards in creating systems with general features, as detailed in Section 3. This paper presents the application of the methodology, the lessons learned from its implementation, and the results of a study evaluating the effectiveness of the final simulator in terms of acquired knowledge, usability, presence, and workload.

The results of this study demonstrated that OPVSim effectively supports students' learning growth, evidenced by significant improvements in their procedural knowledge. This underscores the importance of the methodology employed in developing the final product and highlights the considerable potential of VR to enhance learning experiences in specialized training environments. These findings provide valuable insights for the effective design and implementation of VR systems using existing methodologies. This is crucial for optimizing the creation of such systems and, consequently, improving learning outcomes and facilitating the acquisition of practical skills in complex technical tasks or interactions with machinery.

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We used ChatGPT to edit the English grammar, in order to avoid incorrect translations or writing.

SUPPLEMENTAL MATERIALS

All supplemental materials translated from Spanish to English are available at <https://bit.ly/SupplementalMaterialsOPV>.

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